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In situ analysis of stratified glass eye beads from the tomb of Marquis Yi of the Zeng State in Hubei Province, China using XRF and micro-Raman spectrometry

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A total of 46 stratified glass eye beads in Hubei Provincial Museum, Wuhan, China were analyzed *in situ* by means of portable X-ray fluorescence and Raman spectrometers. These beads were excavated from the famous tomb of Marquis Yi of the Zeng State, which was dated back to the early Warring States Period (433 BC or little late), in Hubei Province. It is determined that most of these elegant glass beads belong to the typical soda-lime-silicate glasses with low contents of MgO and K_2O. The colorants used within these beads are attributed to two types: antimony-based compound opacifiers/colorants and the transition metals in their oxidized state such as Co, Cu, and Fe ions. According to their characteristics of typology, chemical composition, and micro-crystals within samples, we classified them into three subtypes and evaluated the possible relations among them. In addition, through comparing with the similar contemporary samples unearthed from Xu Jialing Tomb (5th–4th century BC) in Henan Province, we discussed the provenance of these stratified glass eye beads and relative issues briefly. Copyright © 2014 John Wiley & Sons, Ltd.

Introduction

As a famous archeological site in the early Warring States Period, the tomb of Marquis Yi of the Zeng State is situated at Leigudun, which is 2 km from the northwest of Suizhou city, Hubei Province. The Zeng State was a minor state subordinate to its powerful neighbor, Chu State. As one of the thousands of tombs dated to the Warring States Period in Hubei with large scale and exact date, the tomb of Marquis Yi has very important research value for archeology and history.[1] Many kinds of antiques have been excavated from this tomb since 1978, including about 200 stratified glass eye beads. Ancient glass is a kind of important artificial materials and evidences to trace the mutual cultural and technical exchanges between East and West.[2] During the past 20 years, only limited glass samples, mainly fragments, from the tomb of Marquis Yi were analyzed, and very few Raman data about these objects have been published.[3–5] The present knowledge of these stratified glass eye beads needs to be further enhanced.

Because most of the glass eye beads from the tomb of Marquis Yi are intact, we need to choose some non-invasive techniques to analyze these beads. Moreover, for the safety of cultural relics, it is better to analyze them *in situ*. Considering the importance of these samples from the tomb of Marquis Yi, we carried on technical analysis using portable X-ray fluorescence (pXRF) and Raman spectrometers.

As a versatile analytical technique, the XRF spectroscopy has been used widely in archeometry because of the distinct features of fast, non-destructive, multi-element analysis.[6,7] In recent years, great advance has been achieved for the portable XRF spectrometry, which is suited well for the *in situ* analysis of artifacts with the artistic value, limited or impracticable remotion, and fragility.[8,9] To know, pXRF spectrometer has been employed to determine nondestructively the chemical composition of ancient glass, ceramics, pigments, metal alloys, and other materials of artistic interest.[10–14] Considering the accurate characterization of the artifacts, the pXRF has been used together with other analysis techniques.[15,16] With the advantages of non-destructiveness, high spatial, and spectral resolutions,[17–19] micro-Raman spectroscopy also becomes one of the most effective tools for the identification of crystalline phases and structural change of ancient glasses, enamels, and mosaics.[20–23]

In the present study, the two complementary analytical techniques allow detail characterization of the chemical composition and micro-crystals within the stratified glass eye beads from the tomb of Marquis Yi. Some valuable information related to the recipe and making technology of these samples were obtained. Combined with the results of archeological research, the possible provenance of these stratified glass eye beads was discussed briefly. This research will shed new light on the glassmaking history and native development of ancient Chinese glass.

Materials and methods

Sample

Forty-six intact stratified glass eye beads were selected for analysis. These artifacts were provided by Hubei Provincial Museum, Wuhan,

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China. All samples were excavated from the tomb of Marquis Yi of the Zeng State dated to the early Warring States Period (433 century BC or little late). These stratified eye beads were opaque or translucent. Figure 1 shows the typical images of these artifacts. In terms of the typology and number, these stratified glass eye beads chosen can be considered to be representative of the glass beads found in this tomb. All intact samples are analyzed *in situ* in Hubei Provincial Museum without any pretreatment prior to experiment. For comparison, the areas with different colors in these beads were measured.

pXRF spectroscopy

The chemical compositions, including some trace elements, of the 46 stratified glass eye beads were quantitatively determined using an OUTSTEX 100FA type portable energy dispersive XRF spectrometer. A palladium (Pd) excitation tube was used (voltage, 40 or 15 kV; current, 0.5–1.0 mA). The X-ray beam spot on the sample had a diameter of about 2.5 mm. The pXRF spectrometer allows the selection of direct, monochromatic, and filter models to detect signals from light and heavy elements. The spectra were acquired for 300 s and split into 100 s for each model. This equipment shows good stability, the relative standard deviation value, and detection limits for most of the major and minor elements in our previous study.[24,25]

Micro-Raman spectroscopy

The vibrational bands of the stratified glass eye beads were nondestructively examined using a LabRam XploRA Raman microspectrometer equipped with a laser source (wavelength, 532 nm; diffraction grating, 1800 g/mm; long-working distance objective, 50×). The laser power was projected onto the focal spot (beam diameter, ~1 μm) of the artifact and maintained at a power of <15 mW. The spectral resolution was <2 cm^{-1} over the spectral range of 50–4000 cm^{-1}. The detection system comprised a multichannel air-cooled charge-coupled device detector. This system was calibrated using the 520.7 cm^{-1} Si line from a single crystal Si wafer at the beginning of each measurement.

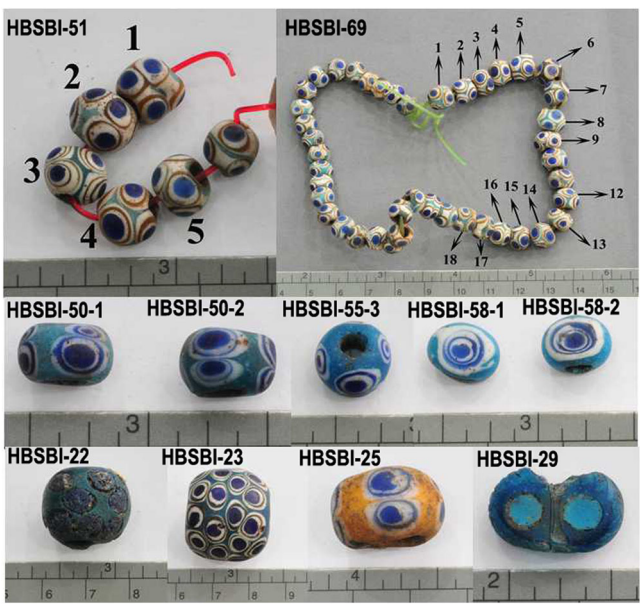

Figure 1. Photographs of the typical analyzed stratified glass eye beads.

Results and discussion

Chemical composition of stratified glass eye beads

Tables 1 and 2 supply the chemical composition (major and minor elements) of 46 beads. Because of surface weathering, the losses of flux sodium are very serious. Except for samples of HBSBI-29 and HBSBI-57-1 (Na_2O, 9.26–13.75%), the weight percentage of Na_2O in all the other samples ranges from 1.69% to 6.44%. The specimens contain 7–13% CaO and 247–528 μg/g Sr. Previously reported data for the fresh surface of one bead fragment (sample HBWKI-18), which were excavated from the same tomb and had similar typology with samples of HBSBI-24 and HBSBI-51, showed that the content of Na_2O was in the range of 13.43–14.58%.[3] All these beads could be attributed to the soda-lime glass. Besides of samples HBSBI-22 and HBSBI-29, the other stratified glass eye beads analyzed are characterized by low contents of MgO and K_2O contents, both less than 1%. We deduce that most of these glass beads from the tomb of Marquis Yi can be categorized under the natron-type soda-lime glass.[26] HBSBI-22 and HBSBI-29 both contain a medium amount of MgO (1.27–5.04%) and K_2O (1.29–2.40%) and are thus assigned to plant-ash type soda-lime glass.[27] It should be mentioned that samples of HBSBI-22, HBSBI-25, and HBSBI-66 have higher PbO content (1.89%, 9.01%, and 2.47%) in their matrix compared with the other beads. The same result goes for HBSBI-55-5 in its white circle (PbO, 1.59%). Although high concentration of minor element Cr was also found in almost all the samples, the reason is still not clear.

Considering the characteristics of chemical composition and typology, these stratified glass eye beads can be classified into three subtypes (Table 3) according to the matrix colors, patterns of eye circles, SnO_2 contents in different parts, and Sb_2O_3 contents in their matrix. Beads of the subtype I have light blue opaque matrices, white and ochre eye circles, and dark blue eyeballs. Twenty-three samples belong to this subtype and do not contain SnO_2 in any part, whereas the content of Sb_2O_3 in their light blue opaque matrix is in the range of 2.00–5.79%. A total of 19 beads of the subtype II have blue-green translucent matrices without ochre circles. The contents of SnO_2 in matrices, white eye circles, and dark blue eyeballs range from 0.01% to 0.64%. The blue-green translucent matrices of those beads belonging to subtype II contain 0.4–2.41% Sb_2O_3. The bronze (slag) could have been used instead of copper. SnO_2 in white is difficult to identify but it may be the impurity of Sb_2O_3.

Four special samples (HBSBI-22, HBSBI-23, HBSBI-25, and HBSBI-29) were attributed to subtype III. They are very different from the former two subtypes in chemical compositions, shapes, and decorations. The HBSBI-22 sample has a blue-green translucent matrix and dark blue eyeball both containing SnO_2 (0.86%, 0.43%). The HBSBI-23 sample with more eyeballs decorated is very similar to subtype II in typology. In addition, SnO_2 exists both in its matrix and white-ochre circles but not in eyeballs. The HBSBI-25 sample has a yellow matrix without SnO_2 in any part. As for HBSBI-29 with special shape, SnO_2 (0.02%) can only be found in light blue eyeballs instead of its blue matrix. It is worth noting that a stratified glass eye bead with similar decorations was excavated from Qiuchengdun Tomb (early Warring States Period) in Hongshan, Jiangsu Province, but no scientific analysis has been reported.[28]

Identification of colorants and opacifiers

Various colorants are used in different areas of the stratified glass eye beads. For example, Cu^{2+} is the main colorant in blue

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Table 1. Chemical composition of major elements determined by pXRF of the stratified glass eye beads from the tomb of Marquis Yi in Hubei Province (wt% oxides normalized to 100)

Samples No.	Subtype	Color of the measured point	Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	K_2O	CaO	TiO_2	MnO_2	Fe_2O_3	CoO	CuO	Sb_2O_3	SnO_2	PbO
HBSBI-21	I	Blue matrix	2.33	0.79	2.61	81.61	n.d.	0.45	7.21	n.d.	0.03	0.54	n.d.	1.56	2.86	n.d.	n.d.
		White circle	2.67	0.45	2.44	78.21	n.d.	0.68	10.16	n.d.	0.03	0.46	n.d.	0.16	4.74	n.d.	n.d.
		Dark blue eyeball	2.09	0.54	2.57	79.20	n.d.	0.51	8.42	n.d.	0.03	2.48	0.36	0.30	3.52	n.d.	n.d.
HBSBI-24	I	Blue matrix	2.82	0.46	2.51	79.41	0.35	0.35	8.56	n.d.	0.02	0.66	n.d.	2.37	2.48	n.d.	n.d.
		White circle	3.64	0.48	1.91	73.38	n.d.	0.15	10.43	n.d.	n.d.	0.30	n.d.	n.d.	9.71	n.d.	n.d.
		Dark blue eyeball	2.21	0.64	2.92	77.31	0.36	0.59	9.33	n.d.	0.02	1.81	0.40	0.48	3.92	n.d.	n.d.
HBSB I-51-1	I	No. 1 white circle	2.42	0.64	0.00	83.17	n.d.	n.d.	7.35	n.d.	0.04	0.30	n.d.	n.d.	6.08	n.d.	n.d.
HBSB I-51-2	I	No. 2 ocher circle	3.29	0.49	1.96	75.52	n.d.	0.67	10.08	n.d.	0.03	0.58	n.d.	0.22	7.17	n.d.	n.d.
HBSB I-51-3	I	No. 4 blue matrix	2.58	0.46	1.05	79.65	n.d.	0.26	8.73	n.d.	0.03	0.37	n.d.	1.08	5.79	n.d.	n.d.
HBSB I-51-4	I	No. 5 dark blue eyeball	2.70	0.42	1.96	75.49	n.d.	0.28	9.57	n.d.	0.03	1.58	0.28	0.18	7.50	n.d.	n.d.
HBSBI-66	I	Blue matrix	3.70	0.58	2.23	75.23	0.00	0.99	9.15	n.d.	0.03	0.48	n.d.	0.98	4.18	n.d.	2.47
	I	Dark blue eyeball	1.69	0.48	2.15	80.29	0.16	0.84	8.04	n.d.	n.d.	2.18	0.29	0.29	3.29	n.d.	0.29
HBSBI-69-1	I	No. 1 dark blue eyeball	2.27	0.80	2.41	75.92	0.42	0.55	9.45	n.d.	0.03	1.84	0.27	0.35	5.68	n.d.	n.d.
HBSBI-69-2	I	No. 2 white circle	2.62	0.59	1.14	80.05	n.d.	0.35	9.28	n.d.	0.06	0.64	n.d.	n.d.	5.26	n.d.	n.d.
HBSBI-69-3	I	No. 3 ocher circle	4.41	0.39	1.95	77.53	0.32	0.14	9.74	n.d.	0.03	0.57	n.d.	0.02	4.89	n.d.	n.d.
HBSBI-69-4	I	No. 4 dark blue eyeball	2.96	0.56	2.20	74.66	0.19	0.19	9.36	n.d.	0.03	2.29	0.40	0.55	6.63	n.d.	n.d.
HBSBI-69-5	I	No. 5 blue matrix	4.39	0.47	1.83	75.05	n.d.	0.73	10.10	n.d.	0.03	0.49	n.d.	1.52	5.22	n.d.	0.18
HBSBI-69-6	I	No. 6 blue matrix	3.54	0.55	2.13	73.80	n.d.	1.60	7.41	n.d.	0.04	6.26	n.d.	2.03	2.00	n.d.	0.65
		No. 6 dark blue eyeball	3.53	0.51	2.99	75.96	n.d.	0.09	8.16	n.d.	0.03	2.01	0.43	0.55	5.74	n.d.	n.d.
HBSBI-69-7	I	No. 7 white circle	4.06	0.54	1.66	73.85	n.d.	0.51	11.45	n.d.	0.03	0.92	n.d.	0.11	6.29	n.d.	0.58
HBSBI-69-8	I	No. 8 dark blue eyeball	2.51	0.53	2.25	74.18	3.36	0.36	10.51	n.d.	0.03	2.01	0.22	0.35	3.67	n.d.	n.d.
HBSBI-69-9	I	No. 9 blue matrix	2.64	0.51	1.64	79.99	0.00	0.51	9.13	n.d.	0.03	0.42	n.d.	1.21	3.93	n.d.	n.d.
HBSBI-69-12	I	No. 12 ocher circle	2.50	0.51	2.33	76.57	1.65	0.72	10.97	n.d.	0.03	0.53	n.d.	0.17	4.02	n.d.	n.d.
HBSBI-69-13	I	No. 13 blue matrix	2.40	0.57	1.63	80.16	n.d.	0.46	8.51	n.d.	0.03	0.47	n.d.	1.42	4.36	n.d.	n.d.
HBSBI-69-14	I	No. 14 dark blue eyeball	2.46	0.71	2.14	78.81	0.43	0.45	8.23	n.d.	0.02	2.25	0.33	0.53	3.64	n.d.	n.d.
HBSBI-69-15	I	No. 15 white circle	2.97	0.64	1.39	77.70	0.21	0.44	10.07	n.d.	0.03	0.77	n.d.	0.08	5.33	n.d.	0.35
HBSBI-69-16	I	No. 16 dark blue eyeball	2.84	0.61	2.30	75.21	0.43	0.65	9.10	n.d.	0.03	2.77	0.40	0.48	5.19	n.d.	n.d.
HBSBI-69-17	I	No. 17 ocher circle	3.04	0.51	1.08	83.42	0.12	0.09	8.14	n.d.	0.05	0.47	n.d.	0.13	2.94	n.d.	n.d.
HBSBI-69-18	I	No. 18 white circle	2.64	0.50	1.07	77.97	n.d.	0.34	10.54	n.d.	0.03	0.55	n.d.	0.34	6.01	n.d.	n.d.
HBSBI-27	II	Blue matrix	2.15	0.63	2.34	82.68	n.d.	n.d.	8.83	n.d.	0.03	0.57	n.d.	1.86	0.72	0.19	n.d.
		White circle	2.25	0.59	3.18	79.47	0.12	0.09	10.17	n.d.	0.02	1.41	n.d.	0.17	2.49	0.04	n.d.
		Dark blue eyeball	2.53	0.45	2.86	78.06	n.d.	0.11	9.43	n.d.	n.d.	2.96	0.81	0.54	2.24	n.d.	n.d.
HBSBI-28	II	Blue matrix	4.43	0.38	0.62	72.17	n.d.	0.02	11.53	n.d.	0.03	2.92	n.d.	5.58	1.09	0.64	0.59
HBSBI-50-1	II	Blue matrix	2.62	0.57	2.96	79.05	0.23	0.44	10.64	n.d.	0.03	0.46	n.d.	1.32	1.37	0.10	0.22
HBSBI-50-2	II	Dark blue eyeball	3.50	0.56	0.94	77.86	n.d.	0.14	8.57	n.d.	0.03	2.76	0.53	1.11	4.01	n.d.	n.d.
HBSBI-55-1	II	Blue matrix	2.14	0.76	1.94	82.33	n.d.	0.08	8.65	n.d.	0.03	0.43	n.d.	2.38	0.73	0.28	0.26
HBSBI-55-2	II	Blue matrix	5.08	0.45	n.d.	83.45	n.d.	n.d.	8.20	n.d.	0.02	0.48	n.d.	1.70	0.42	0.20	n.d.
HBSBI-55-3	II	Blue matrix	1.77	0.68	2.28	82.57	0.03	n.d.	8.21	n.d.	0.03	0.47	n.d.	2.16	0.56	0.14	1.11
HBSBI-55-4	II	Dark blue eyeball	4.98	0.58	2.03	75.22	n.d.	0.32	11.13	n.d.	0.06	1.22	0.84	0.68	2.69	0.01	0.25
HBSBI-55-5	II	White circle	3.11	0.61	1.16	80.36	n.d.	n.d.	8.42	n.d.	0.03	0.96	n.d.	0.56	3.21	n.d.	1.59

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HBSBI-55-6	II	Dark blue eyeball	4.01	0.56	2.05	75.03	0.33	0.26	10.39	n.d.	0.03	3.11	0.60	0.67	2.97	n.d.	n.d.
HBSBI-56	II	White circle	4.84	0.55	1.42	78.88	0.00	0.43	9.49	n.d.	0.03	0.41	n.d.	1.20	2.74	n.d.	n.d.
HBSBI-57-1	II	Dark blue eyeball	9.26	0.50	1.76	70.89	0.06	0.50	11.86	n.d.	0.03	2.41	0.44	0.37	1.92	n.d.	n.d.
HBSBI-57-2	II	Blue matrix	2.19	0.56	2.73	82.59	0.38	0.13	9.20	n.d.	0.03	0.58	n.d.	1.11	0.44	0.06	n.d.
HBSBI-58-1	II	White circle	2.42	0.55	2.03	79.11	0.21	0.45	10.07	n.d.	0.03	0.84	n.d.	0.13	4.17	n.d.	n.d.
HBSBI-58-2	II	Blue matrix	3.02	0.45	2.23	80.35	0.13	0.43	8.94	n.d.	0.15	0.44	n.d.	2.15	1.52	0.19	n.d.
	II	Dark blue eyeball	2.76	0.57	2.59	77.16	0.28	0.67	9.80	n.d.	n.d.	2.27	0.32	0.52	3.05	n.d.	n.d.
HBSBI-60-1	II	Blue matrix	1.98	0.48	1.81	82.85	n.d.	n.d.	8.40	n.d.	0.02	0.38	n.d.	2.24	1.38	0.24	0.22
HBSBI-60-2	II	Dark blue eyeball	5.53	0.50	3.60	73.06	0.08	0.36	10.53	n.d.	0.02	2.42	0.42	0.59	2.89	n.d.	n.d.
HBSBI-67	II	Blue matrix	2.80	0.41	1.01	78.69	n.d.	0.67	8.59	n.d.	0.30	1.29	n.d.	3.50	2.41	n.d.	0.32
HBSBI-68	II	Dark blue eyeball	4.68	0.67	0.99	78.45	n.d.	0.01	8.44	n.d.	n.d.	3.02	0.61	0.52	2.60	n.d.	n.d.
HBSBI-22	III	Blue matrix	5.37	1.74	0.76	70.90	n.d.	2.40	5.39	0.04	0.08	2.19	n.d.	7.25	1.12	0.86	1.89
		Dark blue eyeball	3.02	1.27	1.11	76.76	0.12	1.29	6.58	n.d.	0.06	3.74	0.53	1.29	3.35	0.43	0.45
HBSBI-23	III	Blue matrix	2.75	0.68	2.83	78.27	n.d.	0.67	8.16	n.d.	0.03	0.53	n.d.	3.71	1.76	0.61	n.d.
		White circle	2.58	0.86	3.29	75.70	0.05	0.35	10.50	n.d.	n.d.	0.98	n.d.	0.08	5.53	0.08	n.d.
		Ocher circle	2.66	0.58	3.24	76.07	0.30	0.44	9.71	n.d.	0.03	1.30	n.d.	1.79	3.57	0.32	n.d.
		Dark blue eyeball	2.43	0.53	3.11	76.78	0.34	0.45	8.84	n.d.	0.03	2.46	0.83	0.37	3.83	n.d.	n.d.
HBSBI-25	III	Yellow matrix	4.30	0.44	2.27	72.92	0.28	0.24	8.16	n.d.	0.03	1.04	n.d.	n.d.	1.30	n.d.	9.01
		White circle	6.44	0.41	2.02	72.08	0.15	0.17	10.74	n.d.	n.d.	0.71	n.d.	n.d.	6.89	n.d.	0.40
		Dark blue eyeball	4.27	0.50	2.60	74.66	0.31	0.36	9.97	n.d.	0.02	1.29	0.20	0.13	5.46	n.d.	0.23
HBSBI-29	III	Blue matrix	13.75	5.04	3.22	65.87	n.d.	1.94	7.45	0.10	0.04	0.88	n.d.	1.71	0.00	n.d.	n.d.
		Light blue eyeball	13.45	4.99	3.51	65.18	n.d.	2.24	7.52	0.10	0.03	0.85	n.d.	1.67	0.42	0.02	n.d.

n.d., not detected.

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Table 2. Chemical composition of minor elements determined by pXRF of the stratified glass eye beads from the tomb of Marquis Yi (μg/g)

Samples no.	Subtype	Color of test point	Cr	Ni	Zn	Rb	Sr	Pb	Ba
HBSBI-21	I	Blue matrix	106	4	n.d.	6	242	126	36
		White circle	195	3	19	n.d.	357	475	40
		Dark blue eyeball	112	41	621	n.d.	352	497	24
HBSBI-24	I	Blue matrix	246	n.d.	39	4	378	780	44
		White circle	141	8	75	n.d.	270	279	38
		Dark blue eyeball	184	98	191	4	528	199	36
HBSBI-51-1	I	No. 1 white circle	60	4	n.d.	n.d.	134	125	17
HBSBI-51-2	I	No. 2 ocher circle	172	5	7	8	325	80	36
HBSBI-51-3	I	No. 4 blue matrix	177	n.d.	n.d.	n.d.	228	117	26
HBSBI-51-4	I	No. 5 dark blue eyeball	166	11	309	4	321	131	24
HBSBI-66	I	Blue matrix	180	14	16	n.d.	286	M	21
		Dark blue eyeball	234	21	73	n.d.	329	M	33
HBSBI-69-1	I	No. 1 dark blue eyeball	207	31	116	n.d.	396	525	34
HBSBI-69-2	I	No. 2 white circle	112	3	34	5	272	907	42
HBSBI-69-3	I	No. 3 ocher circle	220	5	38	n.d.	379	268	28
HBSBI-69-4	I	No. 4 dark blue eyeball	172	57	463	n.d.	345	157	19
HBSBI-69-5	I	No. 5 blue matrix	201	n.d.	n.d.	n.d.	375	M	32
HBSBI-69-6-1	I	No. 6 blue matrix	93	51	91	n.d.	258	M	24
HBSBI-69-6-2	I	No. 6 dark blue eyeball	243	81	194	5	381	405	27
HBSBI-69-7	I	No. 7 white circle	181	24	n.d.	n.d.	299	M	30
HBSBI-69-8	I	No. 8 dark blue eyeball	213	21	97	5	508	175	32
HBSBI-69-9	I	No. 9 blue matrix	146	3	n.d.	n.d.	349	1086	38
HBSBI-69-12	I	No. 12 ocher circle	258	10	9	4	461	545	26
HBSBI-69-13	I	No. 13 blue matrix	179	n.d.	6	n.d.	315	306	25
HBSBI-69-14	I	No. 14 dark blue eyeball	208	53	175	7	411	335	39
HBSBI-69-15	I	No. 15 white circle	129	16	50	n.d.	292	M	28
HBSBI-69-16	I	No. 16 dark blue eyeball	221	56	617	n.d.	423	645	43
HBSBI-69-17	I	No. 17 ocher circle	110	n.d.	n.d.	n.d.	287	411	23
HBSBI-69-18	I	No. 18 wite circle	177	8	n.d.	n.d.	338	441	36
HBSBI-27	II	Blue matrix	178	10	n.d.	n.d.	373	867	24
		Dark blue eyeball	168	166	328	n.d.	359	281	28
		White circle	204	98	152	n.d.	461	365	24
HBSBI-28	II	Blue matrix	83	n.d.	134	n.d.	307	M	44
HBSBI-50-1	II	Blue matrix	185	6	21	n.d.	539	M	31
HBSBI-50-2	II	Dark blue eyeball	137	48	138	n.d.	265	188	23
HBSBI-55-1	II	Blue matrix	176	3	8	n.d.	343	M	38
HBSBI-55-2	II	Blue matrix	192	13	4	n.d.	120	230	31
HBSBI-55-3	II	Blue matrix	240	n.d.	45	n.d.	328	M	43
HBSBI-55-4	II	Dark blue eyeball	197	210	67	n.d.	391	M	36
HBSBI-55-5	II	White circle	240	13	111	n.d.	286	M	26
HBSBI-55-6	II	Dark blue eyeball	191	96	186	7	429	262	34
HBSBI-56	II	White circle	143	30	n.d.	n.d.	344	800	31
HBSBI-57-1	II	Dark blue eyeball	198	76	59	11	564	251	28
HBSBI-57-2	II	Blue matrix	234	26	n.d.	n.d.	447	581	40
HBSBI-58-1	II	White circle	185	6	24	n.d.	336	462	32
HBSBI-58-2	II	Blue matrix	200	18	34	n.d.	406	377	40
		Dark blue eyeball	203	71	160	8	438	482	41
HBSBI-60-1	II	Blue matrix	167	n.d.	18	n.d.	348	M	43
HBSBI-60-2	II	Dark blue eyeball	173	65	222	n.d.	435	313	39
HBSBI-67	II	Blue matrix	157	10	92	n.d.	237	M	31
HBSBI-68	II	Dark blue eyeball	131	113	93	n.d.	281	419	27
HBSBI-22	III	Blue matrix	92	n.d.	M	114	n.d.	160	M
		Dark blue eyeball	97	91	M	472	n.d.	237	M
HBSBI-23	III	Blue matrix	158	11	54	n.d.	344	704	39
		Dark blue eyeball	233	233	106	n.d.	518	190	37
		White circle	195	68	76	4	445	276	38
		Ocher circle	207	n.d.	56	10	465	488	36

(Continues)

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Table 2. (Continued)

Samples no.	Subtype	Color of test point	Cr	Ni	Zn	Rb	Sr	Pb	Ba
HBSBI-25	III	Yellow matrix	107	20	60	n.d.	247	M	46
		Dark blue eyeball	159	14	99	8	471	M	41
		White circle	204	n.d.	54	n.d.	372	M	26
HBSBI-29	III	Blue matrix	92	10	n.d.	34	331	758	26
		Light blue eyeball	96	28	n.d.	28	348	351	39

n.d., not detected.

Table 3. Subtype's features of stratified glass eye beads obtained from the tomb of Marquis Yi

Subtype	Colors	Chemical composition	Samples' number
I	They have light blue opaque matrices, white and ochre eye circles, and dark blue eyeballs.	They do not contain SnO_2 in any part, whereas the content of Sb_2O_3 in their light blue opaque matrix is in the range of 2.00–5.79%.	23
II	They have blue-green translucent matrices without ochre circles.	The contents of SnO_2 in matrices, white eye circles and dark blue eyeballs range from 0.01% to 0.64%. The blue-green translucent matrices of those beads belonging to subtype II contain 0.4–2.41% Sb_2O_3.	19
III	They are very different from the former two subtypes in chemical compositions, shapes, and decorations.		4

matrices of the former two subtypes. In the matrices of subtype I, the average content of CuO is 1.45%, and that of Fe_2O_3 is 0.49%. It is obvious that contents of the two components (CuO, 2.4%; Fe_2O_3, 0.8%) in samples of subtype II are higher than those of subtype I. Therefore, the hues of matrices in subtype II are blue-green, whereas those in subtype I are light blue. Meanwhile, the contents of Sb_2O_3 contained in the matrices of two subtypes (subtype I, 1.45–5.79%; subtype II, 0.42–2.41%) are also different. That is the reason why the samples of subtype I look more opaque than those of subtype II do.

For the dark blue eyeballs of all samples, CoO ranges from 0.2% to 0.6% with high Fe level (0.85–3.74%) and low Mn level (0.02–0.06%). Cobalt ions are powerful chromophore agents that should be intentionally designed and added by glass craftsmen. Meanwhile, Zn is observed in dark blue eyeballs of almost all samples from the tomb of Marquis Yi (59–1197 μg/g) and Xu Jialing Tomb (40–250 μg/g),[29] which are both higher than that in blue matrices.

Compared with the blue matrices and dark blue eyeballs, the white circles have relatively lower contents of transition metal elements (e.g. Cu and Fe) but have high Sb_2O_3 (3–6%). Table 4 shows the Raman signatures for 18 beads. The white layers of 12 samples involving subtypes I and III are all characterized by Raman peaks at 234, 320, 333, 521, and 670 cm^{-1}. As presented in Fig. 2(a), the typical spectra of these samples are in good agreement with published values about hexagonal $CaSb_2O_6$.[30] Figure 2(b) shows characteristic Raman bands at 477 and 630 cm^{-1} coexisting with $CaSb_2O_6$ in five beads (subtype II), which result from orthorhombic $Ca_2Sb_2O_7$.[23] As an exception, $Ca_2Sb_2O_7$ is the only crystalline phase observed in HBSBI-55-4. It is proved that the hexagonal $CaSb_2O_6$ can be successfully distinguished from orthorhombic $Ca_2Sb_2O_7$ through Raman microspectroscopy.

Meanwhile, a special phenomenon is observed in HBSBI-25 sample [Fig. 3(a)]. The spectrum of the yellow matrix reveals the most intense scattering peak at 131 cm^{-1}, corresponding to the vibration of Pb_4O tetrahedra in terms of the lattice vibration of the Pb cation. As for two overlapping modes at around 331 cm^{-1} (E_g) and 296 cm^{-1} (F_{2g}), they are attributed to vibrational modes related to Sb–O and Pb–O bonds.[31] The strong A_{1g} mode assigned to the totally symmetric elongation of the SbO_6 octahedra is at about 509 cm^{-1}, and the shoulder at around 456 cm^{-1} may be due to the partial substitution of Sb^{5+} by Zn^{2+}.[32] Comparing with the literature data, we confirmed the presence of bindheimite $Pb_2Sb_2O_7$ in HBSBI-25. The yellow color of the crystals induces yellow opaque glasses with colorless vitreous matrices. This result is consistent with contents of PbO (9.01%) and Sb_2O_3 (1.30%) determined by pXRF. However, the reason why HBSBI-22, HBSBI-55-5, and HBSBI-66 have high PbO contents (1.59–2.47%) is still unknown. As for residue, typical Raman bands for quartz (122, 200, and 460 cm^{-1}) accompanied by $CaSb_2O_6$ are found in the white circles of HBSBI-25 [Fig. 3(b,c)].

Remarks on the technology and provenance of stratified glass eye beads

All over the world, glass eye beads became popular after the reign of Amenhotep III (1391–1353 BC) in the late 18th Dynasty of New Kingdom Egypt. The technique about tabular eye beads composed of layers of glass was rediscovered in the 7th century BC, possibly in Egypt or the eastern Mediterranean.[33] Takashi[34] shows two stratified glass eye beads (14th–13th century BC) similar to HBSBI-29 in shape. At the same time, a great amount of excellent examples of these early beads from Phoenicia (6th–3rd century BC), similar to samples analyzed in this paper, are also exhibited in his book.[34] According to the published data, these beads are usually classified as soda-lime silicate glass. However, native glass eye beads found in China are dated mainly

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Table 4. Raman signatures for stratified glass eye beads obtained from the tomb of Marquis Yi

Samples no.	Subtype	Color	Measured wavenumbers (cm^{-1})	Phases	Chemical formula
HBSBI-24	I	White circle	234, 670	Antimony calcium	$CaSb_2O_6$
HBSBI-51-1	I	Blue matrix	234, 320, 333, 670	Antimony calcium	$CaSb_2O_6$
		White circle	234, 320, 333, 521, 670	Antimony calcium	$CaSb_2O_6$
HBSBI-51-3	I	White circle	670	Antimony calcium	$CaSb_2O_6$
HBSBI-51-4	I	Blue matrix	670	Antimony calcium	$CaSb_2O_6$
		White circle	234, 320, 333, 670	Antimony calcium	$CaSb_2O_6$
HBSBI-69-1	I	White circle	234, 670	Antimony calcium	$CaSb_2O_6$
HBSBI-69-2	I	White circle	234, 670	Antimony calcium	$CaSb_2O_6$
HBSBI-69-3	I	White circle	234, 670	Antimony calcium	$CaSb_2O_6$
HBSBI-69-6	I	White circle	234, 670	Antimony calcium	$CaSb_2O_6$
HBSBI-69-8	I	White circle	234, 670	Antimony calcium	$CaSb_2O_6$
HBSBI-69-13	I	White circle	234, 670	Antimony calcium	$CaSb_2O_6$
HBSBI-55-1	II	White circle	234, 521, 670	Antimony calcium	$CaSb_2O_6$
			477, 630	Antimony calcium	$Ca_2Sb_2O_7$
HBSBI-55-3	II	White circle	234, 320, 333, 670	Antimony calcium	$CaSb_2O_6$
			477	antimony calcium	$Ca_2Sb_2O_7$
HBSBI-55-4	II	White circle	477, 630	Antimony calcium	$Ca_2Sb_2O_7$
HBSBI-55-5	II	White circle	670	Antimony calcium	$CaSb_2O_6$
			477	Antimony calcium	$Ca_2Sb_2O_7$
HBSBI-55-6	II	White circle	233, 670	Antimony calcium	$CaSb_2O_6$
			477	Antimony calcium	$Ca_2Sb_2O_7$
HBSBI-58-1	II	White circle	234, 670	Antimony calcium	$CaSb_2O_6$
HBSBI-23	III	White circle	234, 320, 333, 670	Antimony calcium	$CaSb_2O_6$
HBSBI-25	III	Yellow matrix	131, 296, 331, 456, 509	Bindheimite	$Pb_2Sb_2O_7$
		White circle	122, 200, 460	Quartz	SiO_2
		White circle	234, 320, 333, 670	Antimony calcium	$CaSb_2O_6$

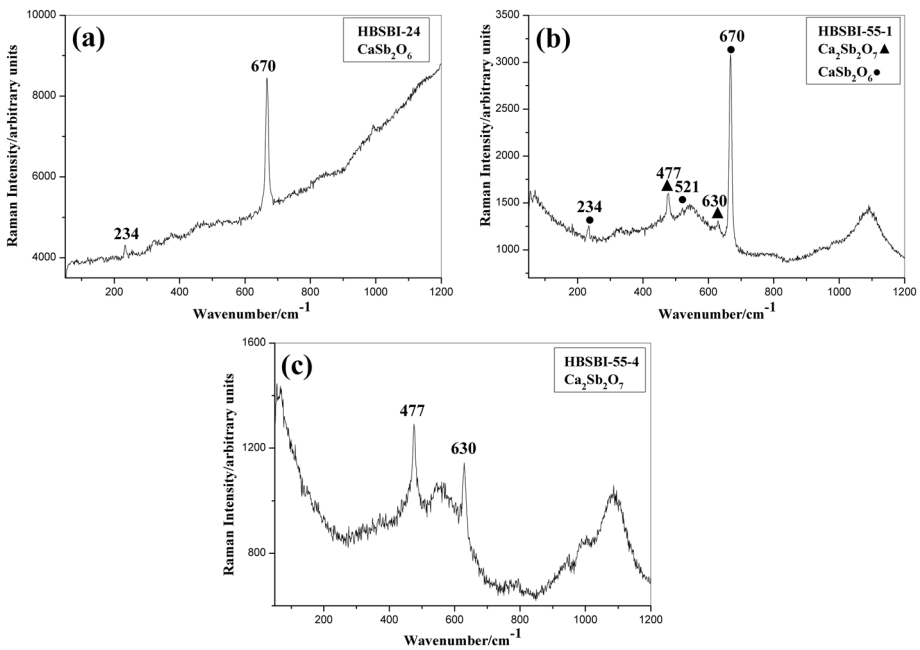

Figure 2. Raman spectra of antimony calcite identified within the white circles of the studied beads. (a) HBSBI-24, white circle, $CaSb_2O_6$. (b) HBSBI-55-1, white circle, $Ca_2Sb_2O_7$ and $CaSb_2O_6$. (c) HBSBI-55-4, white circle, $Ca_2Sb_2O_7$.

from about middle/late Warring States Period to the Western Han Dynasties (4th century BC–1st century AD) and were of lead barium silicate glasses.[2]

Before the 2nd century BC, antimony-based colorants and opacifiers played a predominant role in the production of opaque glasses of Mesopotamia, Egypt, and the Roman Empire.[35,36]

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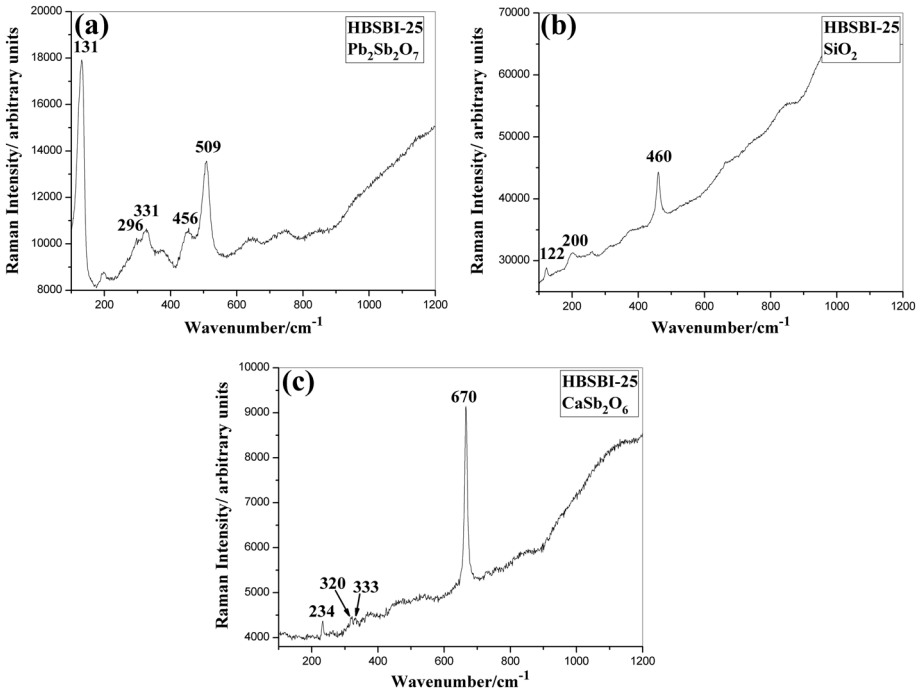

Figure 3. Raman spectra of other crystalline phases identified within HBSBI-25sample. (a) yellow matrix, $Pb_2Sb_2O_7$. (b) white part, SiO_2. (c) white part, $CaSb_2O_6$.

Around the 4th century AD, antimonates seem to have been replaced dramatically in glassmaking by the tin-based colorants and opacifiers in the Byzantine Empire and Europe.[37] Then antimony-based colorants and opacifiers were again used extensively during the Renaissance.[38] Our previous research systematically identified the use of antimony-based colorants and opacifiers in plant-ash type and natron-type soda-lime glass beads (about 1000–400 BC), which are all distributed in the north of the Changjiang River.[5,32] The evidence from these analyzed glasses found in China supports Tite's suggested chronology[37] for the uses of antimony as opacifying agents within the Mediterranean region. This phenomenon may also indicate that before the 4th century BC, the mutual exchanges between China and West related to ancient glass were mainly through the desert and steppe routes of the Silk Road.[2] To our knowledge, no antimony-based opacifiers were identified among the native PbO–BaO–SiO_2, PbO–SiO_2, and K_2O–CaO–SiO_2 glasses in central China from the 5th century BC to the 17th century AD.

The earliest example of cobalt glass colorant is a glass lump dated to 2050 BC, which was excavated at Eridu in Iraq. Cobalt colorant was more widely used by the Egyptian King Tuthmosis III (1479–1425 BC).[39,40] The usage of CoO as colorant for glass in ancient China started from about the Han Dynasties.[4] The native cobalt ores of China were generally known as asbolite,[41] which contains a high percent of MnO, as is analyzed in blue potash glass samples of the Eastern Han Dynasty from Guangzhou.[42] Conversely, for the dark blue eyeballs of our beads, the content of CoO ranges from 0.2% to 0.6% with being low in manganese (0.02–0.06%) and high in iron (0.85–3.74%). They are very similar to the contemporary glass eye beads unearthed from Xu Jialing Tomb[29] (5th–4th century BC) in Henan Province. On the basis of the analytical and archeological evidence, it seems that the cobalt ores used in these glass beads from the tomb of Marquis Yi differ from those of China.

For the introducing ways of antimony-based opacifiers, two hypotheses about *ex situ* and *in situ* crystallization are proposed through synthetic glasses made in the laboratory.[36,43] As $Ca_2Sb_2O_7$ crystals do not exist in nature, the 18th dynasty Egyptian craftsmen may have synthesized $Ca_2Sb_2O_7$ *ex situ* by using mixtures of Sb_2O_3 or Sb_2O_5 with calcium carbonates (atomic ratio Sb/Ca = 1) and treated mixtures between 1000 and 1100 °C.[43] Then glasses are opacified by addition of these crystals $Ca_2Sb_2O_7$ through firing around 1000 °C for 1 h. By contrast, for the blue Roman mosaic tesserae (1st century BC–6th AD), the $CaSb_2O_6$ crystals within the matrices were thought to be made through *in situ* crystallization, using the Sb_2O_4 as an antimony source and undergoing a melting or a heat treatment between 1 and 2 days around 1100 °C.[36] Considering the time sequence of the two hypotheses, we presume that these stratified glass eye beads analyzed here were opacified by *ex situ* crystallization similar to procedures used in Egypt during the second millennium BC. The temperature ranging from 1000 to 1100 °C is plausible for this period.

In brief, combining these pieces of evidence about the typology, chemical composition, colorants, and opacifiers, we prefer to think that these stratified glass eye beads from the tomb of Marquis Yi of Zeng State were imported from the eastern Mediterranean area.

As demonstrated from Tables 1 and 2, the higher contents of MgO and K_2O indicate that the raw materials used to make samples of HBSBI-22 and HBSBI-29, which belong to subtype III, are different from those for the other beads. In addition, we find that HBSBI-22, HBSBI-23, and HBSBI-25 of subtype III take on the characteristics of subtypes I and II in the aspects of typology and chemical composition. Given these aforementioned facts, technologies for beads studied here should be diverse and they should come from different glassmaking centers. The close relationships among the three subtypes imply that there should be mutual exchanges among these centers.

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Conclusions

The complementary use of pXRF and micro-Raman spectroscopy is essential for the *in situ* analysis about 46 stratified glass eye beads from the tomb of Marquis Yi of the Zeng State in Hubei Province. The material used for most bead production is the natron-type soda-lime glass, and they can be classified into three subtypes according to the matrix and eye circle colors, as well as the contents of SnO_2 and Sb_2O_3. Along with the different content of Sb, the transition metals in their oxidized state such as Cu and Fe ions affect the blue hues and the transparency of their matrices. Antimony-based opacifiers/colorants including $CaSb_2O_6$, $Ca_2Sb_2O_7$, and $Pb_2Sb_2O_7$ were characterized from the Raman spectra. According to the decorations, chemical composition, and microcrystals within these beads, the techniques used to make three subtypes should be diverse and they may come from different glassmaking centers with mutual exchanges in the eastern Mediterranean area.

Referring to the soda-lime glass eye beads of the Warring States Period unearthed from China, we think that the desert or steppe route of the Silk Road is the main trade route of glasses between West and East before the 3rd century BC. In the future, in-depth analysis of trace elements using LA-ICP-MS should be performed to provide more information to trace the origin of the glass eye beads from the tomb of Marquis Yi of the Zeng State.

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